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DISABILITY AND REHABILITATION: ASSISTIVE TECHNOLOGY
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RESEARCH ARTICLE

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A systematic review of reminder and guidance systems for Alzheimer's Disease and Related Dementias patients: context, barriers and facilitators

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ABSTRACT

Purpose. To present the latest available research assessing the actual impact of reminder and guidance technologies for daily activities in Alzheimer's Disease and Related Dementia's patients, outlining design implications for these technologies.

Materials and methods. The search was conducted in the ACM Digital Library, IEEExplore, ScienceDirect, PubMed, Cochrane Library, Sage Journal, ResearchGate, and SemanticScholar. An iteratively-developed Boolean search string was built including up to 18 AND/OR terms across Four categories (Memory Aids, Technology, Daily Activities, Memory Impairment). We qualitatively analyzed the findings and discussions of the findings reported in 40 studies in our corpus to determine common barriers to, and facilitators of, effective intervention implementation and adoption.

Results. Forty studies fulfilled the inclusion and exclusion criteria. In existing studies, individuals with Alzheimer's Disease and Related Dementias understand the usefulness of different functions that can be provided within an app to assist with everyday tasks. There was a high level of heterogeneity regarding the studies' location, duration, and evaluation methodology.

Conclusions. There is a need for assistive reminder and guidance technologies to be tailored towards autonomy, identity and personalization. Future work should include motivating features to aid during mood changes and feelings of insecurity.

> IMPLICATIONS FOR REHABILITATION

- In existing studies, older adults with cognitive impairment understand the usefulness of different functions that can be provided within an app to assist with everyday tasks.
- There is a high level of heterogeneity regarding the studies' location, duration, and evaluation methodology.
- There is ample need for these reminder technologies to be tailored towards autonomy, identity and personalization.
- Future work should include motivating features to aid during mood changes and feelings of insecurity.

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Alzheimer; reminder systems; patient-centered design; systematic reviews

Introduction

In our society, human beings' lifespan is increasing, and the average age of our population is getting higher. The elderly population is expected to have rapidly increased by 2050, with the global population of adults in their 60s and 70s doubling (2.1 billion), and individuals over 80 and older predicted to triple (426 million) [1]. As a result, the diagnosis of degenerative illnesses, such as dementia (e.g., Alzheimer's), is also expected to increase. Alzheimer's Disease International (ADI) reports that around 46 million individuals worldwide suffer either from Alzheimer's or other forms of dementia today, and numbers are predicted to reach 131.5 million by 2050 [2].

All types of Dementia, including Alzheimer's, are characterized by a neurologic decline that has a negative impact on the elderly's lives, reducing their quality of life and independence [3]. A range of cognitive problems, such as impaired prospective and episodic memory, and declining thinking and decision-making skills, may result in apathy and difficulties performing basic everyday activities (e.g., eating [3], taking medication [4,5], or getting dressed). Among many other common setbacks, they may fail to recognize their close family members' names and faces [6].

Therefore, individuals with Alzheimer's Disease and Related Dementias (ADRD) need to be looked after and helped by a caregiver to perform their daily activities [7]. Caregivers can be informal or formal. While informal caregivers are usually unpaid family members [3], formal caregivers are paid professionals who take care of their elderly patients [8].

Both formal and informal caregivers face challenges to their own quality of life. In the case of informal caregiving, the demanding nature of looking after relatives or significant others suffering from dementia or any other illness, typically leads to heightened distress and low quality of life, known as caregiver burden. According to previous research, informal caregivers with high burden may experience negative changes in their physical and mental health that may result in high rates of depression [3,9,10]. The level of these caregivers' distress is mostly influenced by the patient, the informal caregiver, and environmental features [11,12]. However, other modulating factors include lengthy hours of care,

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having children, poor perceived health, and feelings of loneliness [11].

Formal caregivers are also likely to experience depression due to the demanding labor of working at nursing homes [8,13]. Their residents' health comes before their own [8,14], which results in excessive caregiving duties [8,15], and physical and emotional exhaustion [8,16]. These factors usually lead to delivering a depersonalized service and having low feelings of professional accomplishment [8,13]. Even though caregiving takes its toll in both formal and informal caregivers, informal ones typically experience higher levels of burden due to much longer care hours, and a shared financial burden with the family member or members they are looking after [17].

Apart from receiving care from formal or informal caregivers, individuals with ADRD could benefit from using assistive technologies, whether as a substitute of paid long-term care services or complementary support [18]. It has been suggested that the use of these technologies by patients could have a positive impact on the caregivers' well-being, whether in informal or formal settings. A study by Anderson et al. [18] suggested that, by using assistive technologies, time and responsibility required by informal caregivers to provide care could be reduced [18]. Nevertheless, in the case of formal caregivers, results are slightly different. According to the authors, even though in most cases the use of these technologies had no impact on care hours, they could still increase (while remaining under the amount of time approved by third-party payers). However, in return, their workload would decrease, reducing any negative outcomes of intensive care [18]. Overall, even in cases where their use could slightly increase care hours, they might still make the caregiver's work more simple [18].

One of the reasons why assistive technologies could have a positive impact on caregivers' well-being is the fact they enable individuals with dementia (PwD) to participate in day-to-day activities. Standard functions include reminding people of activities to perform (e.g., taking medication, mealtimes) or providing instructions to complete complex daily activities (e.g., preparing food recipes) [19]. Reminders, prompts or instructions designed for ADRD care recipients are typically given through digital written notes [20] (although audio, text, picture, and video prompts can be incorporated to support a wide range of tasks [21]), and are usually characterized by their low complexity, portability, ease of use, and for being time-based or context aware [22]. One example of such a digital prompter was supplied by Boyd et al. [21], who explored combining text, audio, and pictures to aid individuals living with dementia through step-by-step tasks. Other functions that have also been proposed to help care individuals with ADRD in their daily living are monitoring devices to keep track of the client's location [23].

By using these technological aids, the care recipients are not only assisted when needed, but their sense of independence and competence is increased. Furthermore their caregivers are relieved from the responsibility of delivering reminders or instructions [22], potentially reducing their burden [24]. However, attitudes towards these technologies continue to present a significant barrier to their uptake [25].

Designing effective systems for individuals with ADRD requires taking into consideration all the needs and challenges faced by this population [26]. Moreover, it is necessary to consider both the caregiver and user's experience with these technologies, and how likely they are to be accepted [25]. Considering this, the lack of definitive guidelines for designing dementia-friendly interfaces is understandable. As the average age of a group increases, so does the diversity of their individual needs and abilities, making a "one size fits all" solution very challenging [26].

To conclude, it is essential to find means to provide effective care, especially to individuals with ADRD, by improving their ability to perform daily living activities, promoting their independence, and increasing their sense of competence. This will help improve the care receivers' behavior regulation skills, and also contribute to a reduced caregiver burden. Unfortunately, studies regarding the design and impact of reminder and guidance technologies for daily activities in ADRD populations remain scarce, especially those aimed at identifying their current barriers and facilitators. Characterizing the use of technology within this population is particularly important given their prevalence, and their potential to reach broader audiences and promote more independent living. Therefore, we conducted a systematic review of papers that focus on technological tools that provide reminders (messages that remind the person to do the activity) and/or daily-activity guidance (instructions of how to do the daily activity) for individuals with ADRD. We believe that our findings will help steer future research on reminders and/or guidance systems for ADRD by characterizing the best practices in terms of design, evaluation, sampling, and the limitations of existing research. Also, they will hopefully help the design of reminder and/or guidance-based technologies solidly grounded in theory that are engaging, personalized, and that promote independence in this population.

This systematic review was guided by four research questions:

- RQ1: How did the context of the patient/target audience inform the design of reminders and/or guidance technologies for daily-living activities in ADRD?
- RQ2: How are reminders and guidance-focused technologies used in ADRD?
- RQ3: What barriers hinder, and what facilitators stimulate, the use of reminders and guidance tools in individuals with ADRD?
- R4: What is the impact of providing reminders and/or guidance technologies in individuals with ADRD?

Method

Our review consisted of six steps: (1) database search, (2) abstract screening, (3) full-text screening, (4) quality assessment, (5) data extraction, and (6) analysis. Given the fast pace the development of methods and technologies have taken in recent years, we examined studies published within the last ten years. This way, we could focus on the relatively more recent approaches.

The papers met the following inclusion criteria: (1) their target population were individuals of any sex and age diagnosed with ADRD; (2) evaluated reminding and/or guiding daily activities for the target population. Papers were excluded whenever they were: (1) published more than ten years ago; (2) published in a language other than English; (3) not focused on ADRD population; (4) not reminding and/or guiding daily activities systems.

Database search

Our database search included articles published between 2012 and 2022. We used eight databases: ACM Digital Library (including specific searches within the SIGCHI database), IEEExplore, ScienceDirect, PubMed, Cochrane Library, Sage Journal, ResearchGate, and SemanticScholar. An iteratively-developed Boolean search string was built including up to 18 AND/OR terms across Four categories (Memory Aids, Technology, Daily Activities, Memory Impairment). The terms in each search varied depending

on the constraints and capabilities of each database. The terms used in the research of articles can be found in Table 1.

Article selection

Using zotero, the initial titles search resulted in 60116 titles out of which 28088 were identified as duplicates, thus remaining 32028 screened articles. The abstracts were analyzed by the authors based on the research questions to obtain the final list of full-text papers to be reviewed. 31913 of them were excluded, as they did not fit the aim of this review. 110 full-text papers were assessed for eligibility and later analyzed to ensure that the studies addressed the actual use of the technology with reminders or instructions of daily activities for individuals with ADRD. We excluded 70 of those, resulting in a final number of 40 papers included for the full systematic review. Figure 1.details PRISMA flow diagram describing the process we followed for this search.

Table 1. Systematic review search keywords.

Memory aids	Technology	Daily activities	Memory impairment
Reminders	Augmented reality	Daily activities	Cognitive impairment
Remember	System	Daily tasks	Dementia
Memory aids	Wearable		Alzheimer
Support remembering	Platform		Alzheimer's disease
			Individuals with Dementia

Quality assessment

We assessed the methodological quality of each article in our corpus using the established study quality assessment tool named Critical Appraisal Skills Programme [27]. Two reviewers assessed each article's quality separately, then reviewed and discussed their findings and any disagreements until they reached an agreement.

Data extraction

Information was extracted from each article based on a predetermined set of data fields, such as the characteristics of the study design and population samples, and the empirical findings (technology outcomes and user experience). The results of all of the studies, the authors' findings, conclusions, and recommendations for future work were extracted for analysis.

Data analysis

We performed a qualitative analysis of the findings reported in the selected studies to determine common barriers and facilitators for the effective adoption and implementation of their proposed interventions. For our analysis, we assigned codes. Therefore, in addition to the barriers and facilitators codes used, we created and assigned them to categories based on said codes. The categories include the common facilitators and barriers identified across the 40 papers in our corpus.

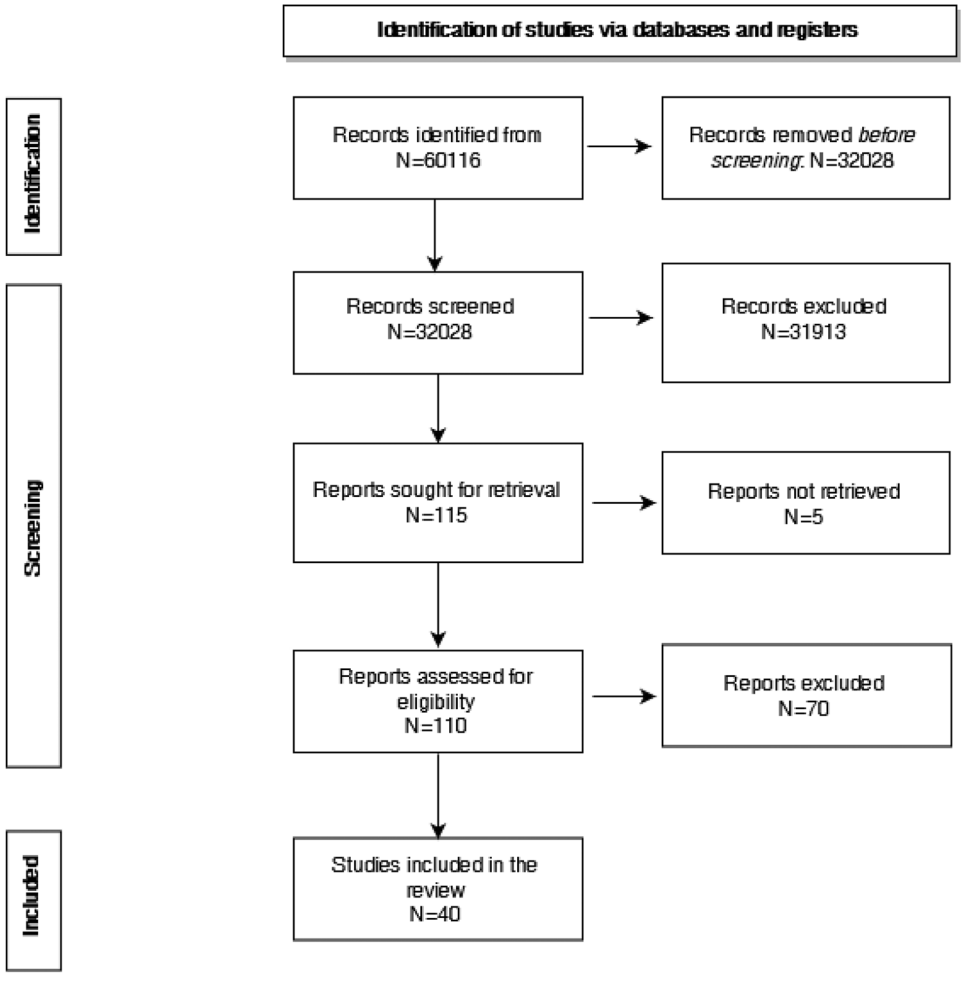


Figure 1. Flow diagram of the systematic literature review search process.

Results

Our analysis of the works represented in our corpus offers an in-depth look into the state of research on technology that helps to remind us of daily activities within ADRD. We first characterize trends in methodology and types of technology explored, reporting on the study design, study population characteristics, and technology characteristics of the projects in our corpus. Through our analysis of study findings and lessons learned, we also characterize barriers to and facilitators of the use of technology in the care of individuals with ADRD, as well as the impact of these interventions. 40 papers are included in our corpus [3–5,7,20,23,28–61].

Study design and measures

Study design and sample size

From the studies in our corpus, 10% were case studies, 10% pilot studies, 5% crossover studies, 3% single case experimental design study, 3% experimental study, 3% user study, 3% survey and in 63% these data were not provided. 85% reported sample size. A total of 591 participants were included in the studies which have an average of 17 participants per study and a range of 1 to 36 participants per study.

3% did more than one study, but did not report the number of participants in all of their evaluations. Because of this, the number of participants of these papers were not included in the total tally. 5% did two studies with the same group of participants. In this case we included the number of participants of each study in the total of participants mentioned above. 18% did preliminary studies during development, which allowed them to implement improvements to the system that helped satisfy the real needs of the target population.

Study location

Out of the selected studies, 60% described the type of setting in which the study was conducted. In these studies, the most reported location was a care facility (18%). The settings of university environments and organizations were sparsely used. Table 2 contains a description of the locations used in these studies. Within the studies that compose this review, 18% were using pre-deployment assistive technologies in a real-world setting outside the lab to better understand the participants' needs and make them part of the development, as mentioned in Meiland et al. [62].

Study duration

In 38% of the studies analyzed in this review that reported duration, there were great variations in the time span of the studies, which ranged from 3 h or less to 3–4 months. However, the duration with more predominance is 3 h or less. In Table 3 a description of the duration used in the studies analyzed is shown.

Type of data

In the studies selected, the types of data used were: 45% qualitative, 35% quantitative and 20% both. Within the qualitative data, semi-interviews represent the majority of collection methods. Qualitative and quantitative data in the same study were sparsely used. Table 4 describes the type of data used in these studies.

Technology evaluation measures

The most reported measure within the studies included in this review was usability, and user satisfaction was sparsely used in usability and user experience. Table 5 contains a description of the type of technological measures used in these studies.

Table 2. Types of locations used in studies.

Authors	Type of Location	Number of studies
Basharudin et al. [30], Jönsson et al. [32], Kosch et al. [38], Lancioni et al. [29,59], Striegl et al. [55], Perilli et al. [58]	Care facilities	17.5% ($n=7$)
Pigot et al. [39], Wang et al. [37], Burleson et al. [54], Begum et al. [57], Hackett et al. [3]	Laboratory environment	12.5% ($n=5$)
Evans et al. [61], Hashim et al. [52], Meinel et al. [35]	Home environment	7.5% ($n=3$)
Boman et al. [46], Tokunaga et al. [53]	Hospital	5% ($n=2$)
Asghar et al. [48], Pirani et al. [7]	Online website	5% ($n=2$)
Kerssens et al. [45]	Hospital or home environment	2.5% ($n=1$)
Rohrbach et al. [4]	Home environment and University of environment	2.5% ($n=1$)
De Jong et al. [40]	Organizations	2.5% ($n=1$)
Baric et al. [51]	University, a library, and meeting room	2.5% ($n=1$)
Law et al. [44]	University and a retirement village	2.5% ($n=1$)
Abu-Dalbouh et al. [50], Alharbi et al. [49], Aljehani et al. [23], Carroll et al. [36], Dewani et al. [41], Ghorbel et al. [20], Harris et al. [47], Kanno et al. [42], Lai et al. [43], McGoldrick et al. [5], Omar et al. [31], Sait et al. [34], Simão et al. [33], Wolf et al. [28], Al-Naami et al. [56], Juárez et al. [60]	Not reported	40% ($n=16$)

Table 3. Duration of the studies.

Authors	Duration of the study	Number of studies
Boman et al. [46], De Jong et al. [40], Lancioni et al. [29], Law et al. [44], Omar et al. [31], Burleson et al. [54], Begum et al. [57], Hackett et al. [3]	less or equal to three hours periods	20% ($n=8$)
Evans et al. [61], Harris et al. [47]	Four weeks	5% ($n=2$)
Jönsson et al. [32], Kosch et al. [38]	Two weeks	5% ($n=2$)
Wang et al. [37]	Four months	2.5% ($n=1$)
McGoldrick et al. [5]	Two months	2.5% ($n=1$)
Kerssens et al. [45]	Three weeks	2.5% ($n=1$)
Abu-Dalbouh et al. [50], Alharbi et al. [49], Aljehani et al. [23], Asghar et al. [48], Baric et al. [51], Basharudin et al. [30], Carroll et al. [36], Dewani et al. [41], Ghorbel et al. [20], Hashim et al. [52], Kanno et al. [42], Lai et al. [43], Pigot et al. [39], Pirani et al. [7], Meinel et al. [35], Rohrbach et al. [4], Sait et al. [34], Simão et al. [33], Tokunaga et al. [53], Wolf et al. [28], Striegl et al. [55] Al-Naami et al. [56], Perilli et al. [58], Lancioni et al. [59], Juárez et al. [60]	Not reported	65% ($n=26$)

Population characteristics

Sex

Sex distribution was calculated for articles that provided data on the proportions of biological male and female participants. Out of the studies analyzed in this review, only 53% reported the participants' biological sex, and women were overrepresented: a total of 89% of the reported participants were females, while the remaining 11% were males.

Table 4. Types of data. User-centered design (UCD) is an iterative design method where designers pay close attention to users' needs at every stage of the process. Thinking aloud is known as the verbalization of thoughts while carrying out an action.

Authors	Type of data	Number of studies
Qualitative data		
Understanding the issues: the opinion of participants experiences		
Asghar et al. [48], Baric et al. [51], Boman et al. [46], De Jong et al. [40], Evans et al. [61], Kerssens et al. [45], Wang et al. [37], Begum et al. [57]	Semi- Interview	20% ($n=8$)
Abu-Dalbouh et al. [50], Alharbi et al. [49], Omar et al. [31], Meinel et al. [35], Sait et al. [34], Juárez et al. [60]	Feedback about the system/ opinion	15% ($n=6$)
Kerssens et al. [45], Kanno et al. [42]	Observations	5% ($n=2$)
Carroll et al. [36]	Usability tests and think-aloud	2.5% ($n=1$)
Wolf et al. [28]	User-centered design approach	2.5% ($n=1$)
Quantitative data		
Opinion of participants to the system		
Aljehani et al. [23], Ghorbel et al. [20], Hashim et al. [52], Pirani et al. [7], Tokunaga et al. [53]	Questionnaire	12.5% ($n=5$)
Quantitative data		
Task- related outcome		
Lancioni et al. [29]	Activities started independently	2.5% ($n=1$)
Lancioni et al. [29,59], Perilli et al. [58]	Activity steps carried out correctly	2.5% ($n=1$)
Lancioni et al. [59]	Duration of the travels	2.5% ($n=1$)
Al-Naami et al. [56]	Fall detection	2.5% ($n=1$)
Dewani et al. [41]	Load testing	2.5% ($n=1$)
Dewani et al. [41]	Peak user load	2.5% ($n=1$)
Lancioni et al. [59]	Travels carried out correctly	3.1% ($n=1$)
Cognitive functioning outcomes		
Harris et al. [47]	ACE-III	2.5% ($n=1$)
Harris et al. [47]	California Verbal Learning Task (CVLT-II)	2.5% ($n=1$)
Harris et al. [47]	Trail Making Component of the Delis Kaplan Executive Functions System Test Battery (D-KEFS-TM)	2.5% ($n=1$)
Caregiver knowledge and attitudes		
Harris et al. [47]	Alzheimer's Disease Knowledge Scale (ADKS)	2.5% ($n=1$)
Harris et al. [47]	Sense of Competence Questionnaire (SCQ)	2.5% ($n=1$)
Ability to perform daily tasks		
Harris et al. [47]	Bristol Activities of Daily Living Scale (BADLS)	2.5% ($n=1$)
Change within rehabilitation research		
Harris et al. [47]	Goal Attainment Scale (GAS)	2.5% ($n=1$)
Qualitative and Quantitative data – Qualitative data		
Understanding the issues: the opinion of participants experiences		
McGoldrick et al. [5] Hackett et al. [3], Burleson et al. [54]	Questionnaire	7.5% ($n=3$)
Basharudin et al. [30], Kosch et al. [38]	Observation	5% ($n=2$)
Jönsson et al. [32]	Focus group	2.5% ($n=1$)
Striegl et al. [55]	Semi-interview	2.5% ($n=1$)
Task-related outcomes		
Basharudin et al.	Chi-square test	2.5% ($n=1$)
Kosch et al. [38]	Cooking performance	2.5% ($n=1$)
McGoldrick et al. [5]	Frequencies were calculated for a percentage of target events remembered out of all events each week	2.5% ($n=1$)
Jönsson et al. [32]	How many times a participant independently came to the dining room	2.5% ($n=1$)
Hackett et al. [3]	Prospective memory task	2.5% ($n=1$)
Basharudin et al. [30]	T-Test	2.5% ($n=1$)
Burleson et al. [54]	Dressing errors	2.5% ($n=1$)
Striegl et al. [55]	Questionnaire	2.5% ($n=1$)
Caregiver outcome		
Jönsson et al. [32]	Caregiver opinion about care technology	2.5% ($n=1$)

Age

The average age was calculated from papers that provided the mean age. Variance statistics were inconsistently reported, so we only calculated the average ages. In the studies selected, 60% reported actual age, however 16% reported a range of age. Due to this, only 44% studies were included, and the weighted mean age of participants with ADRD was 78, of elderly participants without ADRD was 73, informal caregivers 69, formal caregivers 41, and of other participants 22.

Participants

Among the studies analyzed in this review, 93% of papers describe the type of participants. In these studies, 25% reported informal caregivers as participants. In the studies with informal caregivers there were always other participants as well. Formal caregivers with other participants, individuals without disabilities or individuals with ADRD in combination with other participants appear sparsely. Table 6 contains details on the type of participants included in these studies.

Technology characteristics

Population-targeted design

The objective of all studies analyzed in this review was for individuals with ADRD to retain the ability to perform ADL autonomously.

Their technology choice was mainly based on the fact of being promising and the increase in its use. Other important factors were the participants' familiarity with the technology, its usability, and whether the system was designed considering the target user's limitations. Furthermore, the technology efficacy and cost-effectiveness was taken into account, considering it is generally used to overcome the patient and/or caregiver's difficulties.

Finally, technology selection was also based on whether it had already been produced and had been developed by a partner in their research for older individuals. In the studies selected, 15% of the interventions were also customized. The reasons for this customization were the user population's characteristics (50%), technology familiarization (33%), and the population's difficulties with daily activities (16.6%).

Table 5. Types of measures.

Authors	Type of technology measures	Number of studies
Usability and User experience		
Alharbi et al. [49], Basharudin et al. [30], Carroll et al. [36], Kosch et al. [38], Lai et al. [43], Lancioni et al. [29,59], McGoldrick et al. [5], Omar et al. [31], Pirani et al. [7], Meinel et al. [35], Tokunaga et al. [53], Wang et al. [37] Burleson et al. [54], Striegl et al. [55], Al-Naami et al. [56], Begum et al. [57], Perilli et al. [58], Juárez et al. [60], Hackett et al. [3], Alharbi et al. [49], Ghorbel et al. [20], Pirani et al. [7], De Jong et al. [40], Jönsson et al. [32], Pigot et al. [39]	Usability (technology reaction, achieved the main project target, easy to use, user – friendly)	65% ($n = 26$)
Aljehani et al. [23], Evans et al. [61], Harris et al. [47], Kosch et al. [38], Ghorbel et al. [20], Kanno et al. [42], Lai et al. [43], Meinel et al. [35], Abu-Dalbouh et al. [50], Begum et al. [57]	User experience (acceptance, attitude, engagement)	25% ($n = 10$)
Abu-Dalbouh et al. [50], Alharbi et al. [49], Law et al. [44], Omar et al. [31], Begum et al. [57], Baric et al. [51]	Usefulness	15 % ($n = 6$)
Begum et al. [57], Hackett et al. [3]	Efficacy	5% ($n = 2$)
Hashim et al. [52], Pigot et al. [39]	User's satisfaction	5% ($n = 2$)
Feasibility and Sustainability		
Aljehani et al. [23], Asghar et al. [48], Boman et al. [46], Omar et al. [31], Sait et al. [34], Simão et al. [33]	Functionalities and usage	15% ($n = 6$)
Tokunaga et al. [53], Wolf et al. [28]	Feability of the concept	5% ($n = 2$)
Dewani et al. [41]	Sustainability of application	2.5% ($n = 1$)

Type of technology

Within the selected studies, the most commonly assessed device was smartphone device, corresponding to 18% of our corpus. The remaining devices assessed included computer combined with other technology and wearables. Table 7 shows the types of technology used in these studies.

Features for providing reminders or instructions

Out of all the studies of our corpus, 63% provided reminders for daily activities. Using both reminders and instructions was uncommon. More specifically, 28% used audio reminders such as alarms, notifications, warnings, and beeps; 13% used combined modalities such as visual and auditory reminders; and 10% used visual reminders such as text messages. Table 8 there is detailed information about the types of reminders used in these studies.

Within the 28% of studies that provided instructions, the most reported modalities were visual and audio directions. Visual-only guidance was the least used. Table 9 contains detailed information about the type of instructions used in these studies.

Table 6. Types of participants.

Authors	Type of participants	Number of studies
Informal caregivers with other participants		
Evans et al. [61], Harris et al. [47], Kerssens et al. [45], Begum et al. [57], Hackett et al. [3]	Informal caregivers and individuals with Dementia	12.5% ($n = 5$)
Aljehani et al. [23], Wang et al. [37]	Informal caregivers and participants with Alzheimer Disease	5% ($n = 2$)
Lai et al. [43]	Informal caregivers and participants with Dementia or Cognitive Impairment	2.5% ($n = 1$)
Kanno et al. [42]	Informal, formal caregivers and individuals who are older	2.5% ($n = 1$)
Carroll et al. [36]	Informal caregivers and healthcare care providers of individuals who are older	2.5% ($n = 1$)
Elderly or ADRD participants		
Ghorbel et al. [20], Lancioni et al. [29,59], McGoldrick et al. [5], Rohrbach et al. [4], Perilli et al. [58]	Participants with Alzheimer Disease	15% ($n = 6$)
Hashim et al. [52], Wolf et al. [28]	Participants with Cognitive Disease	5% ($n = 2$)
Meinel et al. [35]	Elderly participants	2.5% ($n = 1$)
Formal caregivers with other participants		
Kosch et al. [38]	Formal caregivers and participants with Cognitive Impairment	2.5% ($n = 1$)
Striegl et al. [55]	Formal caregivers and elderly participants	2.5% ($n = 1$)
De Jong et al. [40]	Formal caregivers and experts	2.5% ($n = 1$)
Formal caregivers with other participants		
Sait et al. [34]	Formal caregiver, family members,participants with Dementia, doctor, implementation partners and designer	2.5% ($n = 1$)
Jönsson et al. [32]	Formal caregivers, Managers and participants with Dementia	2.5% ($n = 1$)
Tokunaga et al. [53]	Formal caregivers and participants with Dementia	2.5% ($n = 1$)
Individuals without disabilities		
Burleson et al. [54], Bassam Al-Naami et al. [56]	Healthy participants	5% ($n = 2$)
Pigot et al. [39]	Graduate student and worker	2.5% ($n = 1$)
Baric et al. [51]	Senior population	2.5% ($n = 1$)
Alharbi et al. [49]	Students	2.5% ($n = 1$)
Omar et al. [31]	Under-graduate student and faculty member	2.5% ($n = 1$)
ADRD participants with other participants		
Abu-Dalbouh et al. [50], Basharudin et al. [30], Simão et al. [33]	Participants with Alzheimer and participants who are older	7.5% ($n = 3$)
Boman et al. [46]	Participants with Cognitive Impairment, health care professionals and significant other	2.5% ($n = 1$)
Law et al. [44]	Participants with Cognitive Impairment and experts in aged care	2.5% ($n = 1$)
Asghar et al. [48]	Participants with Dementia and volunteers	2.5% ($n = 1$)
Dewani et al. [41], Pirani et al. [7], Juárez et al. [60]	Not reported	7.5% ($n = 3$)

Table 7. Types of technology.

Authors	Type of technology	Number of studies
Mobile device		
Abu-Dalbouh et al. [50], Alharbi et al. [49], Asghar et al. [48], Dewani et al. [41], Hashim et al. [52], Pirani et al. [7], Hackett et al. [3]	Smartphone device	17.5% ($n=7$)
Boman et al. [46], Evans et al. [61], Harris et al. [47], Jönsson et al. [32], Lai et al. [43], Tokunaga et al. [53]	Touchscreen Tablet	15% ($n=6$)
Mobile device combined with other technology		
Aljehani et al. [23], Sait et al. [34]	Mobile device and smart watch	5% ($n=2$)
Baric et al. [51]	Mobile device and a computer or tablet	2.5% ($n=1$)
Omar et al. [31]	Mobile device and smart pill box with LCD screen and buzzer	2.5% ($n=1$)
McGoldrick et al. [5]	Mobile device and tablet	2.5% ($n=1$)
Kanno et al. [42]	Mobile device and a tracker	2.5% ($n=1$)
Carroll et al. [36],	Mobile device and web application	2.5% ($n=1$)
Burleson et al. [54]	Tablet and wearable t-shirt	2.5% ($n=1$)
Lancioni et al. [29]	Tablet and wireless bluetooth earpiece	2.5% ($n=1$)
Robots		
Law et al. [44], Rohrbach et al. [4], Simão et al. [33], Wang et al. [37], Striegl et al. [55], Begum et al. [57]	Robot	15% ($n=6$)
Computer and web application		
Lancioni et al. [59]	Computer	2.5% ($n=1$)
De Jong et al. [40]	Computer monitor	2.5% ($n=1$)
Kerssens et al. [45]	Touchscreen computer	2.5% ($n=1$)
Ghorbel et al. [20]	Web application	2.5% ($n=1$)
Ambient sensors		
Pigot et al. [39]	Ambient assisted living system with sensors	2.5% ($n=1$)
Basharudin et al. [30]	Red and green ambient lights	2.5% ($n=1$)
Augmented Reality		
Wolf et al. [28]	Augmented Reality Head – Mounted Display	2.5% ($n=1$)
Kosch et al. [38]	Projection based system with kinect and speaker	2.5% ($n=1$)
Television combined with other technology		
Meinel et al. [35]	Smart TV and smart watch	2.5% ($n=1$)
Juárez et al. [60]	TV and medicine kit interface	2.5% ($n=1$)
Computer combined with other technology		
Perilli et al. [58]	Computer, speakers, optic sensors and light-reflecting paper	2.5% ($n=1$)
Wearables		
Al-Naami et al. [56]	Wireless-sensing smart wearable medical device	2.5% ($n=1$)

Within the 8% of studies that provided both reminders and instructions, the most used modalities were audio reminders and visual instructions combined. Verbal reminders and instructions were scarcely used. Table 10 shows the type of instructions and reminders used in these studies. Additionally, in 13% of the studies, the caregiver was able to select which modality to use, therefore choosing their preferred type of reminders or instructions. This allowed the caregivers to choose the modality that would

Table 8. Types of reminders.

Authors	Type of reminders	Number of studies
Audio reminders		
Boman et al. [46], De Jong et al. [40], Kanno et al. [42], Kerssens et al. [45], Pigot et al. [39]	Voice	12.5% ($n=5$)
Hashim et al. [52], Al-Naami et al. [56]	Alarm	5% ($n=2$)
Sait et al. [34]	Alarms and notifications	2.5% ($n=1$)
Ghorbel et al. [20]	Beep	2.5% ($n=1$)
Omar et al. [31]	Buzzer sound	2.5% ($n=1$)
Pirani et al. [7]	Pop-up and notification	2.5% ($n=1$)
Visual reminders		
McGoldrick et al. [5], Meinel et al. [35], Baric et al. [51]	Text message	7.5% ($n=3$)
Juárez et al. [60]	Icons and text	2.5% ($n=1$)
Visual and Audio reminders		
Hackett et al. [3]	Auditory alarm sound and text	2.5% ($n=1$)
Jönsson et al. [32]	Music and icons accompanied short text	2.5% ($n=1$)
Tokunaga et al. [53]	Voice, text and images	2.5% ($n=1$)
Other combination of modalities		
Evans et al. [61]	Text, audio and picture	2.5% ($n=1$)
Simão et al. [33]	Visual , Auditory and Olfactory cues	2.5% ($n=1$)
Abu-Dalbouh et al. [50], Alharbi et al. [49], Aljehani et al. [23], Dewani et al. [41], Lai et al. [43]	Not reported	12.5% ($n=5$)

Table 9. Types of instructions.

Authors	Type of instructions	Number of studies
Visual and audio instructions		
Kosch et al. [38]	Contour projections , video instructions and voice instructions	2.5% ($n=1$)
Wolf et al. [28]	Green guidance cue and voice	2.5% ($n=1$)
Harris et al. [47]	Verbal, text or photographs	2.5% ($n=1$)
Wang et al. [37]	Video clip and voice	2.5% ($n=1$)
Begum et al. [57]	Video and voice	2.5% ($n=1$)
Rohrbach et al. [4]	Voice including a displayed subtitle and a holographic simulation	2.5% ($n=1$)
Visual instructions		
Basharudin et al. [30]	Ambient light cues	2.5% ($n=1$)
Burleson et al. [54]	Image and text	2.5% ($n=1$)
Lancioni et al. [59]	Photos	2.5% ($n=1$)
Perilli et al. [58]	Video Prompting, Pictorial Cues	2.5% ($n=1$)
Law et al. [44]	Not reported	2.5% ($n=1$)

Table 10. Types of instructions.

Authors	Type of reminders and instructions	Number of studies
Verbal reminders and Verbal instructions		
Lancioni et al. [29], Striegl et al. [55]	Verbal reminders and Verbal instructions	5% ($n=2$)
Audio Reminders and Visual instructions		
Carroll et al. [36]	Voice reminders and Proactive assistant	2.5% ($n=1$)
Asghar et al. [48]	Not reported	2.5% ($n=1$)

be best for the patient. The most common customized feature was that of reminders, with 8% of studies enabling reminder type adaptation, 5% providing customization of instruction types, and 3% enabling the adaptation of both.

Regarding the use of reminders and/or instructions, only 13% reported that further information was delivered to the caregiver if the patient missed the reminder or did not perform the indicated actions. These cases were few, as most of the studies in our corpus did not mention delivering complementary information to the caregiver.

Other features

Within the studies that compose this review, 39% reported having more features other than reminders or instructions. The features that were mostly provided were patient tracking (GPS), photo and name identification, or information to help the patients review or construct their memories. Other features like video calls and wake up the ADRD patient were less prevalent. Table 11 shows detailed information about other features used in these studies.

Table 11. Other features.

Authors	Other features	Number of studies
Alharbi et al. [49], Aljehani et al. [23], Dewani et al. [41], Kanno et al. [42], Omar et al. [31], Sait et al. [34], Al-Naami et al. [56]	Tracking patient (GPS)	17.5% ($n=7$)
Alharbi et al. [49], Ghorbel et al. [20], Kanno et al. [42], Pirani et al. [7],	Photo and names identification or information of family or friends	10% ($n=4$)
Boman et al. [46], Law et al. [44], Meinel et al. [35]	Emergency call or Emergency help	7.5% ($n=3$)
Aljehani et al. [23], Omar et al. [31], Al-Naami et al. [56]	Heart rate signals or stress	7.5% ($n=3$)
Ghorbel et al. [20], Hashim et al. [52], Pirani et al. [7],	Memory game	7.5% ($n=3$)
Boman et al. [46], Pigot et al. [39]	Contacts	5% ($n=2$)
Lancioni et al. [29], Wolf et al. [28]	Motivation	5% ($n=2$)
Asghar et al. [48], De Jong et al. [40]	News	5% ($n=2$)
Dewani et al. [41], Pirani et al. [7]	Quiz games	5% ($n=2$)
Asghar et al. [48], Pigot et al. [39]	Weather information	5% ($n=2$)
Asghar et al. [48]	Call	2.5% ($n=1$)
Boman et al. [46]	Care plan	2.5% ($n=1$)
Al-Naami et al. [56]	Fall detection	2.5% ($n=1$)
Pirani et al. [7]	Help me / SOS Button	2.5% ($n=1$)
Juárez et al. [60]	Locator for keys, wallets and other things	2.5% ($n=1$)
Asghar et al. [48]	Messages	2.5% ($n=1$)
Boman et al. [46]	Monitoring	2.5% ($n=1$)
Law et al. [44]	Mood check	2.5% ($n=1$)
Sait et al. [34]	Object identification	2.5% ($n=1$)
Asghar et al. [48]	Painting memories	2.5% ($n=1$)
Meinel et al. [35]	Process of disease and the health status	2.5% ($n=1$)
Pirani et al. [7]	Treatment and rehabilitation center	2.5% ($n=1$)
Asghar et al. [48]	Travel tutor	2.5% ($n=1$)
De Jong et al. [40]	Video call	2.5% ($n=1$)
Law et al. [44]	Wake up the ADRD patient	2.5% ($n=1$)
Abu-Dalbouh et al. [50], Baric et al. [51], Basharudin et al. [30], Carroll et al. [36] , Evans et al. [61], Harris et al. [47], Jönsson et al. [32], Kerssens et al. [45], Kosch et al. [38], Lai et al. [43], McGoldrick et al. [5], Rohrbach et al. [4], Simão et al. [33], Tokunaga et al. [53], Wang et al. [37], Hackett et al. [3], Lancioni et al. [59], Perilli et al. [58], Begum et al. [57], Striegl et al. [55], Burleson et al. [54]	Not reported	52.5% ($n=21$)

Technology use: barriers and facilitators

Patients – low interest/motivation and familiarization

In some studies (45%), the participants were generally positive towards the new technology, which was a facilitator of intervention success. However, other authors reported different results.

Wang et al. [37] reported that, though open to the idea, the participants were not prepared for robot assistance. They further reported a mismatch in opinions between care receivers with mild-to-moderate AD and their caregivers regarding robot use, adding that both the caregivers' and receivers' preferences should be considered before recommending this type of devices. In Lai et al.'s study [43], most of their participants with cognitive impairment had low interest in trying the proposed app, despite receiving a demonstration of its use. This lack of interest points to a barrier in technology adoption.

Tokunaga et al. [53] reported that, whether the subjects can use the contents on screen-based devices (e.g., smartphones or tablets), depends strongly on their familiarity with them. In their population with very few technological-literate participants, many had problems interacting with the device. Regardless, most of them became more familiar with these interfaces as the experiment progressed, and a few were used to smartphones and had no difficulties.

Rohrbach et al. [4], reported that their patients were not prepared for the weight of bulky wearable devices, needing to rest from them depending on intervention length. The most prevalent issue reported was the discomfort of wearing the gear due to its physical characteristics (e.g., size, bulkiness, weight, obstructiveness), followed by their lack of motivation to learn how to use new technology because of their advanced age.

Patients – remember daily activities

Most of the reviewed studies (80%) found that reminders and step-by-step guidance helped complete tasks and guide patients to perform a variety of tasks correctly.

For instance, Basharudin et al. [30] found that ambient cues provided the users guidance to correctly perform the task regardless of the time taken. Tokunaga et al. [53] report that instructions with pictures were easier to understand than those without pictures.

A study by Boman et al. [46] reported that the most important functionality of their proposal was a shared Calendar that could be accessed by SOs (Significant Others) and FCs (Formal Caregivers), most likely for facilitating task planning and setting timed reminders.

Patients – engaging content

Some studies (30%) reported that their participants enjoyed using technology, which included visual, audio, and text cues, to perform activities of daily living. As mentioned in McGoldrick et al. [5], the participants enjoyed the autonomy and independence the tested app provided. Regarding their overall impression of the

app, a participant mentioned that she liked the "idea of it, highlighting how the app" gives him something for himself… "a sense of independence", instead of receiving instructions on everything.

However, a lack of engagement was reported by Evans et al. [61]. In their study, some participants (both caregivers and care receivers) perceived technology as too problematic to use, which was attributed to the impairment of cognitive functioning caused by both aging and dementia. In other cases, their lack of interest in technology reflected a personal lifestyle choice.

Patients – acceptability

Participants in some studies (42.5%) clearly accepted technology, and were more likely to have positive outcomes.

Hashim et al. [52] reported an effectiveness of over 70%, showing the high acceptance of the application as a potential source of help. Reportedly, the fact that the app fit into their existing patterns of need led to such positive outcomes. Kanno et al. [42] further claimed that using verbal alarms as a reminder and assisting in locating medication was very much accepted by their participants.

However, other authors reported different results. In Rohrbach et al.'s study [4], some patients recognized the potential of technology to help them with their daily tasks (especially the cueing system). Regardless, others were unwilling to use the application, felt the experience as too novel and unusual, did not fully grasp its concept, or were questioning the application's usefulness. This might be the result of the participants' denial about their diagnosis or thought of not needing any support at home.

Lai et al.'s [43] findings suggest that individuals with cognitive impairment appeared to have a great desire to maintain their independence, and prefer on-call caregiver assistance over a constant presence. They were open to the usefulness of the reminder function, but less open to the idea of their carer being able to update their appointment information remotely.

Regarding the acceptance of robots, Simão et al. [33] reported that the robot used was well received. On the other hand, in Wang et al. [37], none of the elderly individuals expressed the immediate desire to use one, even though they were open to the idea. Their results suggest a lack of perceived need for the robot's services, which might grow as symptoms start to threaten the patient's independence [62].

Patients – usefulness

Most reviewed studies (67.5%) reported that their participants considered the tools provided useful. Hashim et al. [52] claimed that their tool enhanced the individuals' reminiscence, stimulated their cognitive function, and helped them perform their daily activities. Kosch et al. [38] reported that auditory instructions were preferred over video and contour to convey and process information. However, though not always deemed needed, video and contour instructions were considered more useful for understanding details.

Patients – emotional impact

In a few studies (32.5%), participants reported positive feelings (e.g., motivation, confidence, and happiness) from using technology.

In Hashim et al.'s study [52], the patients reportedly felt motivated and comfortable using their application. The app's multimedia elements, touch screen feature and error-free simple interface facilitated system navigation, and increased the patient's confidence to explore the application without feeling nervous or uneasy.

Kerssens et al. [45] reported that most of their participants indicated that the application helped them relax, enjoy life, and reminisce; it neither made them feel monitored nor did they find it irritating. Jönsson et al. [32], found one of their participants noticeably happier and more confident using the application than before its introduction. However, they clarified that the fact the application was presented as a tool to promote independent task completion might have influenced this response.

Other studies associated using these technologies with feelings of awareness of their own limitations. McGoldrick et al. [5] mentioned how technology was interpreted by participants as a reminder of what they could not do, which led to anxiety or other negative responses. The authors reported the participants' frustration when errors occurred, as they expected a fully functional system. Their participants also considered the reminder app did not take into account other memory difficulties they were experiencing.

Sometimes, the use of technology carried possible stigma, as exemplified in Evans et al.'s study [61], where many participants felt that the need of being prompted reflected their lack of competence to complete everyday living tasks. This led to feelings of embarrassment and social stigma in some participants.

Informal or formal caregivers – familiarization, acceptability and usefulness

A few studies (37.5%) reported feedback from informal or formal caregivers. In general, the caregivers' attitude towards using technology, including assistive technology, to help in daily-living activities, was positive.

Informal caregivers in Aljehani et al.'s study [23] described the application presented as user-friendly. Kerssens et al. [45] reported that the proposed technology, not only made the informal caregivers' daily lives easier but also facilitated the task of helping their care receiver. Furthermore, most of them claimed they would miss using the technological companion. However, they also reported that while assistive technologies can help mitigate symptoms and address needs in PWD, care burden and other challenges remained.

On a less positive note, as reported by Evans et al. [61], some informal carers struggled to understand the purpose of the prompter, and how relevant it would be to their lifestyle and routines. As a result, they indicated lack of confidence in using it.

In Wang et al.'s study [37], informal caregivers identified instrumental activities of daily living (e.g., cooking and cleaning) to help the elderly with, as they often wished to contribute to household activities. They also highlighted the need for different levels of assistance, from high-level reminders to stepwise prompts, as many activities required detailed step-by-step prompting to avoid uninitiated, missed, or out of sequence steps.

Jönsson et al. [32] reported that formal caregivers working at the facility where their study took place generally had a positive attitude towards trying new technology, as long as they got a sense of what it was about and how it may be beneficial. The same authors highlighted that their graphical user interface design and using tablets were positive for the usability of the system by formal caregivers. They expressed that limiting the apps' features to reminders and notifications about meals might have influenced this positive outcome. Regardless, the need for a more robust support system was reported.

Informal caregivers – emotional impact

Not many studies (13.3%) reported the emotional impact of their interventions on caregivers, and the few studies that did had informal caregivers as participants. Wang et al. [37] identified that having robots could lighten up informal caregivers' workload and allow them to focus on more valued activities with more individuals with ADRD.

The stigma of using technology in informal caregivers was mentioned in few studies (13.3%). Evans et al. [61] reported that, for their participants, using prompters had to be weighed against their fear of undermining their partner's self-esteem. Their difficulties using the technology, and the effort that took to persuade their loved one with dementia to use it could also cause tension and negatively affect their relationship.

Outcomes

We aimed at identifying how successful reminder and/or guidance-based assistive technology has been at improving memory and performance of daily activities of individuals with ADRD. To do this, we examined the impact on life outcomes reported by studies focused on the aftermath of these technologies use. These studies were chosen given that the effects of technology are widely considered the goal standard for experimentally measuring its impact in daily life.

Our findings showed a positive impact of these technologies on daily living. Most studies (80%) yielded at least mildly significant improvements in the evaluated measures, including support for everyday functions and increased effective interaction between the users and their environment. This subsequently increased the users' independence, self-confidence and self-esteem.

Discussion

This systematic review describes the work done on the designing and evaluating technologies that provide reminders and/or guidance of daily activities for individuals with ADRD. Due to the increasing number of individuals suffering from ADRD, there is a growing number of research examining how technology can support them in the performance of their daily activities. An increasing subset of this work has explored opportunities for innovation and possible improvements in reminders and/or instructions of daily activities for individuals with ADRD.

While our review identified many papers using these tools, there have been few contributions that allow the customization of reminders and instructions. Furthermore, by characterizing the barriers and facilitators to the adoption of technology and its success across previous work, we highlight promising directions for the design of future applications. We now turn to a discussion of important considerations for future research.

Sex

Within the studies that compose this review, the predominant sex was female, presumably because women have higher incidence of dementia than men, with dementia rates diverging at about age 85, and Alzheimer's Disease (AD) rates at around 80. Women tend to live longer than men, possibly due to different etiologies, preferential male attrition and lower thresholds of disease pathology needed to produce symptoms, as mentioned in Beam et al. [63]. For future work, we encourage recruiting more male participants to have a more balanced sample in the studies.

Location of the reviewed studies

The context in which an individual interacts with the reminders or the guidance influences how these technologies should be designed and helps to contextualize why such tools are adopted (or not). Nevertheless, 40% of the studies analyzed did not report the implementation location(s).This was unexpected, given that their focus was on technologies that helped participant with ADRD retain the ability to perform activities of daily living autonomously. By definition, providing reminders and/or guidance to the patients' daily activities and/or their caregivers is meant to support them while at home. Implementing these technologies in hospitals or a laboratories could bias the results. As stated in [62], testing in a real-world setting, outside the laboratory, is important when developing this kind of technology. It contributes to a better understanding of the users' needs, includes the participants in the development process, and ensures a good match between the person and the application or device. If this is not achieved from the end user's perspective, the technology may be ignored or not be used optimally.

We encourage future research to take place in the participant's home, and report the setting used and its implications on technology delivery in much more detail. This will help understand the technology's potential through the participants' reactions in a real-world context.

Duration of studies

Out of the studies in this corpus, 63% did not report their intervention length. This information could be important to better understand the effectiveness of the various implementations, taking into account that longer studies are more likely to result in more effective and culturally appropriate interventions. In the remaining 38% of the studies, there were great variations in intervention length, ranging from three hours or less to four months.

Additionally, 15% did report having done any pre-intervention work (development stages) to inform their design and implementation. Extending the interaction with participants through pre-intervention work may help researchers avoid making premature judgments and jumping to conclusions. Having multiple development stages may give authors a more nuanced understanding of the population under study.

Future work should report the study settings, length and, if possible, carry out pre-intervention studies with the target population, from the onset of the idea to its final implementation.

Technology evaluation measures

We identified significant barriers to comparing findings across studies. First, there were high variations in the way intervention outcomes, such as usability, technology satisfaction, and participants' behavior, were measured.

The most common measures used were focused on usability. Testing usability is important since it allows researchers to determine how easy it is for participants to use technology. This feedback is essential for making technology as user-friendly as possible, defining what works well and what needs improvement [64].

Other common measure used was that of user experience. These included participants' behaviors and attitudes toward the proposed technology. However, most of these studies only measured their user's experience at the end of the study. Although this may be appropriate for short studies, it can also limit our knowledge on how the participants' experience evolves.

In comparison, assessing the users' satisfaction with technology is lacking. Measuring user satisfaction would help researchers to find out what participants like and dislike about the proposed applications, what leads to their dissatisfaction, and identify the participants' specific needs [65].

Overall, future work on assistive technology in this field should include usability and user satisfaction measures. We also encourage researchers to use standardized measures and tools consistently across studies about reminders and/or guidance for daily activities. This would facilitate future comparisons, which would in turn result in more robust evaluations.

Participants

40% of the studies included informal or formal caregivers, 25% only informal caregivers and 15% formal caregivers and other participants (e.g., participants with ADRD). Including informal and formal caregivers is important, as they can give feedback on whether a solution should be adopted further or halted. It is reasonable to think that it is important for informal caregivers to feel that their opinions matter and that they have an impact on the decisions made for their love ones. For caregivers to adopt a system, factors such as familiarity, usefulness, engagement or lack of interest, acceptance, and the emotional impact of the proposed technology are important. Also, according to the literature reviewed in this paper, participants who suffer from ADRD have rarely been involved in the development of the technologies that are meant to support them.

We encourage the inclusion of caregivers, especially informal ones, and participants with ADRD in feedback sessions to ensure that both understand the assistive technologies proposed, give opportunities to find the answers to their questions, and collaboratively discuss recommendations for improvement. To promote a more effective use of assistive technologies in ADRD care, understanding their usability for the end users, their level of acceptance, and people's preferences for specific interfaces is key. Furthermore, it is also important for informal caregivers to understand these technologies, as this could help to reduce their burden and care better for their loved ones.

Technology and features to provide reminders, instructions or both

The articles analyzed in our corpus highlighted several ways these reminder and/or guidance-based tools for daily activities could better engage users. Mobile applications are amongst the most commonly used, and they have shown potential in providing reminders while increasing the users' sense of independence, potentially supporting individuals with ADRD.

Similarly, according to a review by Asghar et al. [66], recent developments in digital solutions have the potential to support individuals with aphasia. However, these authors noted, more work is required from both the government and academic institutions to research and fund the rehabilitation of those who have this condition.

Feature of motivation

Among the papers in our corpus, only 5% mentioned any features aimed to motivate the target population. This was an unexpected finding, since individuals with ADRD tend to experience feelings of insecurity, low self-esteem, and even depression symptoms [67]. Thus, they are prone to experience uneasiness about using new technology, requiring additional motivation to maintain their functional engagement over time [67]. Adding features with the potential to increase motivation would help design a more accessible system for the target audience.

Summary of considerations for future studies

This article shows what and which elements future studies should take into consideration when carrying out studies focused on reminders and/or guidance of daily activities for individuals with ADRD. Future studies should have the following factors into consideration:

Study location – home environment

Studying these technologies in the context where they will be used provides a better understanding of their potential. It provides a more accurate picture of the users' needs and makes the participants part of the development process. It also helps forge a better match between individual and technology. If this match does not exist from the end user's perspective, the proposed technology may be ignored or not used optimally.

Duration – longer and several stages

By conducting longer studies, such as four-week ones, with several stages where participants could give feedback early throughout development, the researchers would have a more nuanced understanding of the population under study. This would, in turn, result in more effective and culturally appropriate interventions.

Technology measures – usability and user satisfaction

Using measures applied to technology, such as usability and user satisfaction, would help researchers to understand the participant's relationship with the proposed devices and applications, what works well, what needs improvement, and what participants like and dislike. Furthermore, they allow us to identify the participants' specific needs more easily.

Participants – caregivers and individuals with ADRD

Including caregivers, whether formal or informal, and individuals with ADRD ensures that the end users understand the assistive technologies proposed, are able to ask/answer questions, and that they collaboratively discuss recommendations for improvement. It is important to take into account that technology can be used by many individuals with ADRD, but additional support is often needed from informal caregivers or professionals.

To promote a more effective use of assistive technologies in ADRD care, we need to better understand their usability for individuals with ADRD, as well as the individual's preferences and acceptance for specific technologies and interfaces.

Type of technology – mobile

Using mobile technologies will give individuals with ADRD a positive sense of independence.

Feature – customization

Individual variations in needs and abilities mean "one size fits all" is not an optimal solution. The adaptability of the technology to the users' needs must be considered if we want to increase acceptance with regards to technology.

Feature – motivation

Designing more accessible systems would help mitigate the impact of apathy, mood changes, feelings of insecurity and difficulties with initiation, common symptoms of neurodegenerative disease [67]. Furthermore, having motivational features like badges and rewards would motivate the participants to maintain their functional engagement over time.

Limitations and strengths

In this article, we present an overview of studies on the topic of reminder and guidance systems for ADRD patients, showing that in existing studies, people with ADRD understand the usefulness of different functions that can be provided within an app to assist with everyday tasks. We show that there is ample need for these reminders and guidance systems to be tailored towards autonomy, identity, and customization. Future work should include motivating features to aid during mood changes and feelings of insecurity.

Only studies published in English within the last ten years (from 2012 to 2022) featured in eight databases were included. Therefore, this research does not take into account papers published earlier or written in other languages

Conclusion

Individuals suffering from ADRD have shown to be enthusiastic about technology aimed at remembering daily activities and task guidance. Such technology has allowed them to remain independent, and they are able to take part in the design process of emerging technologies and applications. In the reviewed studies, these individuals have shown to understand the usefulness of different functions provided within an app to assist with everyday tasks. However, the need for assistive technologies to be tailored towards autonomy and identity maintenance was still apparent. Aligning assistive technology design to these needs is key, as it may improve this population's technology uptake.

Some other challenges still remain. Among them, developing strategies that facilitate customization of these applications, choosing the best type of reminder or guidance for each user, and helping the caregivers to select the best solutions for their patients or family members.

We envision that involving the end-users in the development process of new technologies and applications with reminders and/or guidance of daily activities will continue to grow. Further, we foresee that more applications of existing technologies using mobile phones or apps will be used for the benefit of individuals with ADRD.

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